When Will a Borrower Default?

Survival Analysis in Credit Risk

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Should these be the same outcome?

- One borrower defaults after three months.
- Another defaults after two years.

Both defaulted. The timing is different.

Hypothetical loans.

A third borrower has not defaulted yet

Observed for six months

No default so far

What happens after month six is not observed.

A binary label needs a horizon

- A classifier can predict default within 12 months.
- Six months without default does not establish a 12-month label.

Training and evaluation labels need appropriate follow-up.

Which 12-month labels are known?

Optional explanation

Observed history	Default by month 12?
Default at month 3	Yes
Default at month 24	No
No default through month 6	Unknown

Hypothetical loans; no earlier competing exit in these histories.

Why survival analysis?

- Use when events occur.
- Accommodate right-censored follow-up under suitable assumptions.
- Here: Cox for interpretable relative default hazards.

Machine-learning methods can also model survival outcomes.

Choose the event and the clock

Origination $\longrightarrow$ first recorded default

Elapsed time since the loan was issued

Follow-up can end in different ways

ILLUSTRATIVE · Hypothetical loans; months since origination



C: right censoring — active when observation ends.

D: closure — a competing event ending default exposure.

The survival function

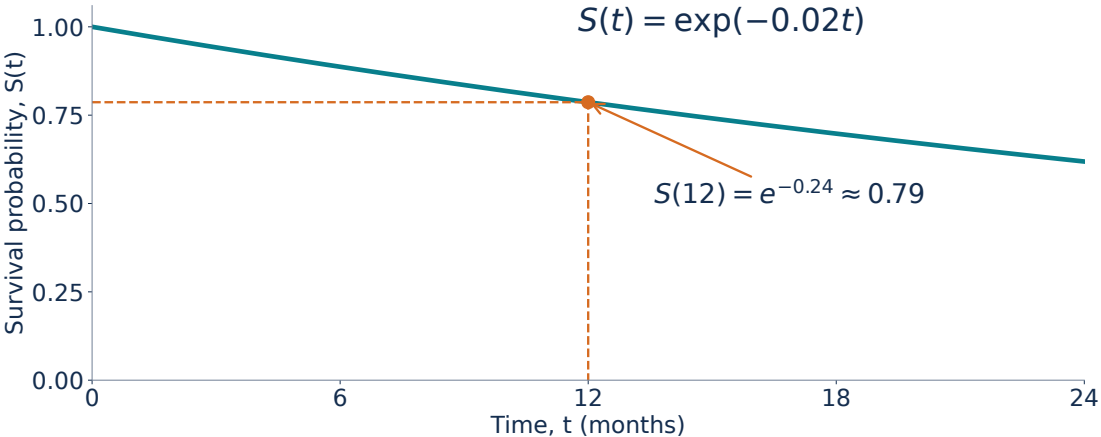
First use a hypothetical default-only setting.

$$S(t) = P(T_D > t)$$

The probability of remaining default-free beyond time t.

Survival: the probability of remaining default-free

HYPOTHETICAL · Default-only setting; not a Bondora estimate



At 12 months, about 79% remain default-free in this hypothetical setting.

How rapidly are defaults occurring now?

Among loans still exposed now:

how rapidly do defaults occur next?

Hazard: an instantaneous rate, not a probability.

From a short interval to a hazard rate

Hypothetical default-only setting; t in months.

$$h(t) = \lim_{\Delta t \downarrow 0} \frac{P(t \le T_D < t + \Delta t \mid T_D \ge t)}{\Delta t}$$

$$P(t \le T_D < t + \Delta t \mid T_D \ge t) \;\approx\; h(t)\,\Delta t$$

Rate $\times$ a short interval $\approx$ a conditional probability.

From hazard to survival

Optional explanation Default-only setting.

$$H(t) = \int_0^t h(u)\,du$$

$$S(t) = \exp\{-H(t)\}$$

With competing closure, default hazard alone does not determine actual default probabilities.

Borrowers have different characteristics

Age Income Customer status

Can these be associated with different default rates?

The Cox model

$$h(t \mid x) = \underbrace{h_0(t)}_{\text{baseline}} \underbrace{\exp(\beta^{\mathsf{T}} x)}_{\text{multiplier}}$$

Baseline: the time pattern for reference covariates.

Multiplier: how the covariates scale that rate.

For credit: a cause-specific default rate among loans still exposed.

Compare otherwise identical borrowers

Hypothetical new vs existing customer; same baseline and other covariates.

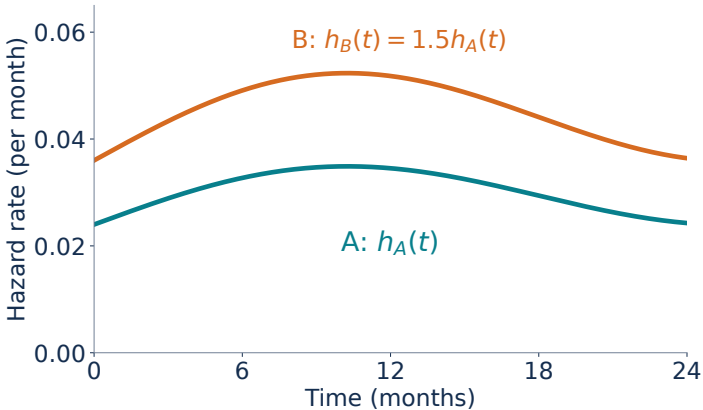
$$\mathrm{HR} = \frac{h(t \mid \text{new})}{h(t \mid \text{existing})} = \exp(\beta_{\text{new}})$$

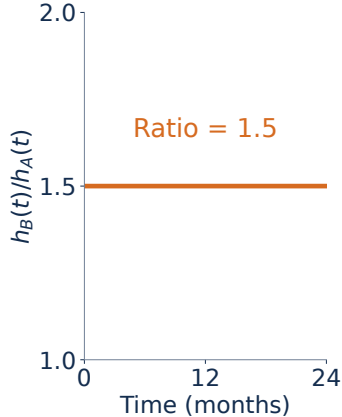
If $\mathrm{HR} = 1.5$: a 50% higher rate at each t, under PH.

A conditional rate ratio, not a probability ratio or causal effect.

Proportional hazards: changing rates, constant ratio

HYPOTHETICAL · A hazard is an instantaneous rate, not a probability





The baseline hazard can vary over time; proportionality concerns the ratio.

How does Cox learn relative effects?

Illustrative: three borrowers at risk just before one default.

A	B	C
Existing customer	Existing customer	New customer
Relative risk 1	Relative risk 1	Relative risk e^{β}

Which borrower accounts for this default?

Cox regression compares borrowers in the risk set

ILLUSTRATIVE · At risk immediately before a default; all other covariates equal

A	B	C
Existing customer	Existing customer	New customer
Relative risk: 1	Relative risk: 1	Relative risk: e^{β}

$$P(C \text{ defaults} \mid \text{one default in risk set}) = \frac{e^{\beta}}{2 + e^{\beta}}$$

If $\beta = \log(2)$: $\quad \frac{2}{1+1+2} = \frac{1}{2}$

Conditional on one default occurring now; not a probability over a time horizon.

The audited Bondora cohort

Estonia · Issued 2013–2015

Follow-up: at most 730 days from origination

9,844	6,924	2,508
Loans	Borrowers	Recorded defaults

Reporting cutoff: 20 July 2021; all loans have 730 days of potential follow-up.

Follow-up uses audited endpoints

- Default: first recorded `DefaultDate`.
- Closure proxy: `ContractEndDate` only when snapshot `Status=Repaid`.
- Stop at the first exit or 730 days.

Closure is not guaranteed full cash repayment. The source default threshold is unverified.

Five predictors and their contrasts

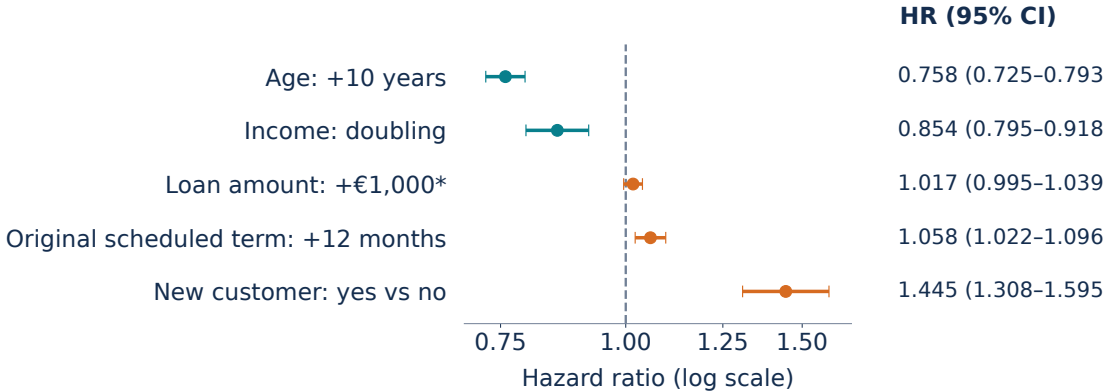
Age	+10 years
Income	Doubling
Amount	+1,000 recorded monetary units
Original scheduled term	+12 months
New-customer status	Yes vs no

Year-stratified baseline; borrower-clustered intervals.

Monthly original-schedule proxy; source currency/income period unverified.

Recorded default: adjusted hazard ratios

EMPIRICAL · Final audited Cox estimates; baseline stratified by origination year



Borrower-clustered 95% CIs are approximate with tied daily event times.

PH departures: age and new-customer status. Nonlinearity: age; borderline for term.

*Monetary fields as recorded; source currency/period definitions remain unverified.

Conditional associations, not causal effects or default probabilities. See REPORT.md.

New versus existing customer

$$\text{HR} = 1.445$$

95% CI: 1.308–1.595

About 44% higher conditional default rate among loans still exposed.

Association, not a default-probability ratio or causal effect.

PH concern: this single HR is a model summary.

Where interpretation is limited

- PH concerns: age and new-customer status.
- Nonlinearity; approximate tied-date intervals.
- Recorded endpoints and measurement proxies.

Back to the lending question

- Three months and two years have different timing information.
- Six default-free months are useful, incomplete follow-up.
- Cox links covariates to conditional default rates.

Questions

Backup: partial likelihood

$$L(\beta) = \prod_{j:\,\text{default}} \frac{\exp(\beta^{\mathsf{T}} x_{i_j})}{\sum_{k \in \mathcal{R}_{y_j}(t_j)} \exp(\beta^{\mathsf{T}} x_k)}$$

$\mathcal{R}_{y_j}(t_j)$: loans still exposed immediately before t_j, within the defaulting loan's origination-year stratum.

Schematic formula for distinct event times. The audited fit uses Efron handling of ties.

Backup: corrected diagnostics

	PH rank p	PH log p	Nonlinear p
Age	0.00641	0.00768	0.00358
Log income	0.933	0.621	0.185
Amount	0.236	0.118	0.805
Original term	0.777	0.997	0.0438
New customer	0.00888	0.000417	—

PH: approximate independence-based Schoenfeld tests, not cluster-adjusted. Rank uses elapsed event times. Nonlinearity: existing spline cluster-Wald checks; exploratory.

One loan per borrower: 6,924 loans, 1,842 defaults. Broadly similar coefficient directions; sample composition changes.